Student's Index No.: _____ Signature: _____

UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER TWO EXAMINATIONS, 2024/2025

TIME: 2 HOURS

COURSE CODE: **TEEG 214**

COURSE TITLE: **EARLY GRADE SCIENCE II**

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and three from Section B.

SECTION A

[25 marks]

Answer all the questions in this section.

1. What is the role of chlorophyll in photosynthesis?
   a) It absorbs sunlight and converts it into chemical energy.
   b) It provides structural support to plant cells.
   c) It helps plants retain water.
   d) It transports nutrients within the plant.

2. Which of the following processes occurs in plants because of sunlight exposure?
   a) Cellular respiration
   b) Transpiration
   c) Photosynthesis
   d) Germination

3. What is the primary purpose of photosynthesis in plants?
   a) To generate heat energy
   b) To release carbon dioxide
   c) To produce glucose and oxygen
   d) To break down organic matter

4. What is the primary purpose of food in the human body?
   a) To provide shelter and protection
   b) To supply essential nutrients and energy
   c) To support cognitive development
   d) To aid in social interactions

5. Food serves as a source of essential nutrients. Which nutrient is primarily responsible for repairing tissues and building muscles?
   a) Vitamins
   b) Carbohydrates
   c) Proteins
   d) Fats

6. Which category of food is rich in carbohydrates and serves as the body's primary source of energy?
   a) Food for growth
   b) Energy-given food
   c) Protective food
   d) Junk food

7. What is the function of a resistor in an electronic circuit?
   a) To store electrical energy
   b) To amplify signals
   c) To control the flow of electrical current
   d) To generate radio waves

8. Which electronic component is primarily used to store electrical charge and release it when needed?
   a) Diode
   b) Capacitor
   c) Transistor
   d) Inductor

9. The absorption and utilization of food by the body is fundamental to_____.
   A. digestion
   B. nutrition
   C. plants
   D. photosynthesis

10. There are _____ basic groups of food.
   A. four
   B. one
   C. three
   D. two

11. The foods that increase body tissues resulting in the growth and repair of these body tissues are referred to as _____.
   A. balance diet foods
   B. body building foods
   C. energy giving foods
   D. protective foods

12. Eating balanced diets provide the following to the body except _____.
   A. cause illnesses
   B. prevent infections
   C. prevent illnesses
   D. provide the body with energy to work

13. Pick the odd out of the following _____.
   A. carbohydrates
   B. fats
   C. proteins
   D. vitamins

14. The following are electronic components except _____.
   A. capacitor
   B. cell
   C. resistor
   D. transistor

15. The fresh edible portions of certain herbaceous plants are _____.
   A. beans
   B. fruits
   C. grains
   D. vegetables

16. The special coloured chemical that traps sunlight for plants during photosynthesis is called _____.
   A. Chlorophyll
   B. leaf
   C. pigment
   D. stomata

17. Plants prepare or make their food through a special process known as _____.
   A. cooking
   B. nutrition
   C. respiration
   D. photosynthesis

18. Which tool is designed for cutting various materials, such as paper, fabric, and cardboard?
   A. Screwdriver
   B. Pliers
   C. Shears
   D. Hammer

19. When would you typically use a tool like a crowbar?
   A. To tighten screws
   B. To cut wires
   C. To remove nails or pry open objects
   D. To measure lengths

20. What is the primary function of a screwdriver in mechanical work?
   A. Cutting materials
   B. Gripping objects
   C. Driving screws into surfaces
   D. Measuring angles

21. The portion of the electromagnetic radiation given off by the Sun, in particular infrared, visible, and ultraviolet light is called _____.

22. The by-product of photosynthesis is _____.

23. The primary function of a screwdriver in mechanical work is _____.

24. A wheelbarrow is commonly used for _____.

25. Pliers are often used for _____.

SECTION B

[75 marks]

Answer three questions from this section.

1. a) Briefly explain personal hygiene to a basic three (3) child. **[3 Marks]**
   b) State the aim of personal hygiene. **[3 Marks]**
   c) Explain how economic determinants such as cost, income, and availability influence food choices. **[6 Marks]**
   d) List three (3) electronic components and state their functions. **[9 marks]**
   e) State four (4) uses of sunlight to humans **[4 Marks]**

2. a) State **two** (2) health benefits and **two** (2) effects of sunlight to humans. **[4 Marks]**
   b) Explain the term photosynthesis. **[5 Marks].**
   c) How will you distinguish visible light from Infrared radiation **[6 Marks]**
   d) Distinguish between food substances and food nutrients. **[4 Marks]**
   e) Write short notes on
      i. body building foods
      ii. energy giving foods **[6 Marks]**

3. (a) Explain why sunlight is essential for most plants' growth and survival. What specific processes in plants depend on sunlight? **[10 Marks]**
   (b) State three (3) importance each of observing the following personal hygiene:
      a. Cleaning the teeth
      b. Washing the hand
      c. Bathing **[9 Marks]**
   (c) State the uses of the following simple machines:
      1. Hammer
      2. Pincers
      3. Screwdriver
      4. Wheelbarrow
      5. Bottle opener
      6. Scissors **[6 Marks]**

4. (a) Explain the diverse roles that food plays in our lives, focusing on its uses and the various categories of food. Include examples to illustrate your points. **[10 Marks]**
   (b) List five (5) uses of energy from the sun. **[5 Marks]**
   (c) List and explain briefly the function or role of any five (5) electronic components in an electronic device/ circuit. **[10 Marks]**

5. (a) Describe the significance of simple electronic components in the operation of electronic gadgets and toys. Provide examples of electronic components, explain their functions, and discuss how they contribute to the functionality of common electronic devices and toys. **[10 Marks]**
   (b) State and explain any three (3) ways of ensuring personal hygiene. **[9 Marks]**
   (c) Define what body building foods are and state three (3) examples of such group of food. **[6 Marks]**